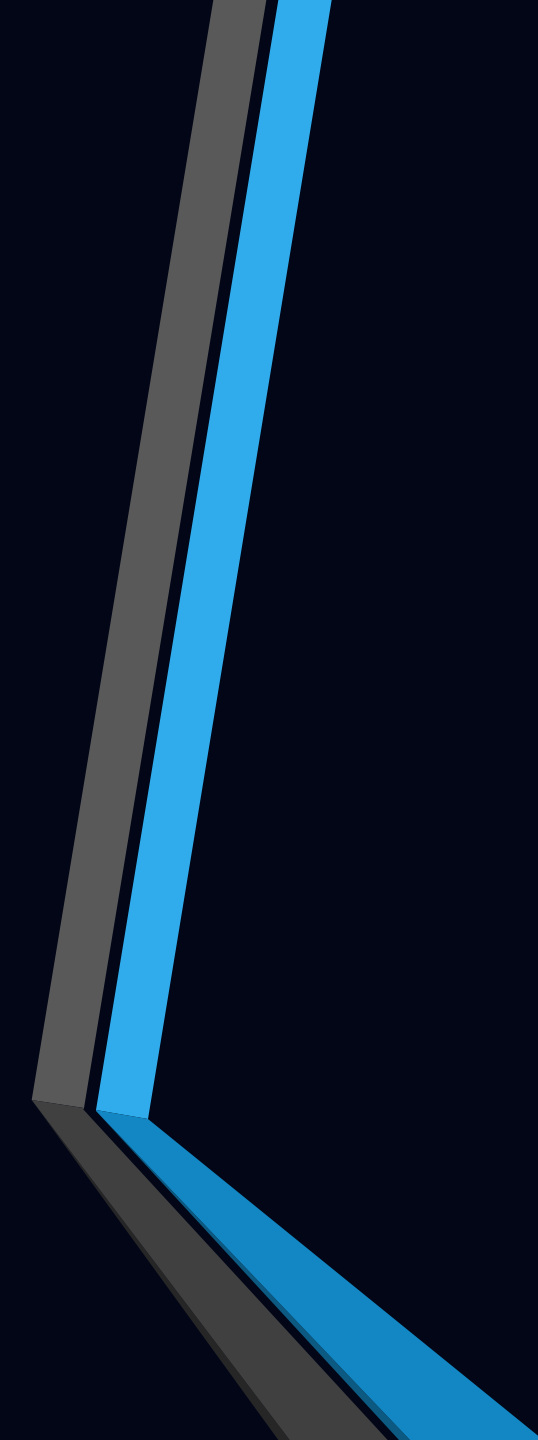




eScanX

PDF Document Analyzer

Intelligent document scanning and OCR processing

Lingaya's Vidyapeeth • Bachelor of Technology • 2025-2026

Project Team

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Under Guidance of

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Problem Statement

- Inability to search text within scanned PDFs
- Manual effort required for document analysis
- Incorrect document orientation affects OCR
- Lack of selective page-level processing

Project Objectives

- Develop a desktop application for PDF analysis
- Support both searchable and scanned PDFs
- Integrate OCR for image-based documents
- Enable keyword search with page-level precision
- Create a professional, user-friendly interface

Key Features of eScanX PDF Document Analyzer

- ✓ Text extraction from searchable PDFs
- ✓ OCR-based processing of scanned PDFs
- ✓ Automatic page rotation correction
- ✓ Selective page range scanning
- ✓ High-accuracy keyword search functionality
- ✓ Generation of searchable PDF outputs
- ✓ Real-time, color-coded runtime logs
- ✓ Professional Graphite-themed UI

System Architecture



Technology Stack



Front-End:

- .NET WinForms (C#)

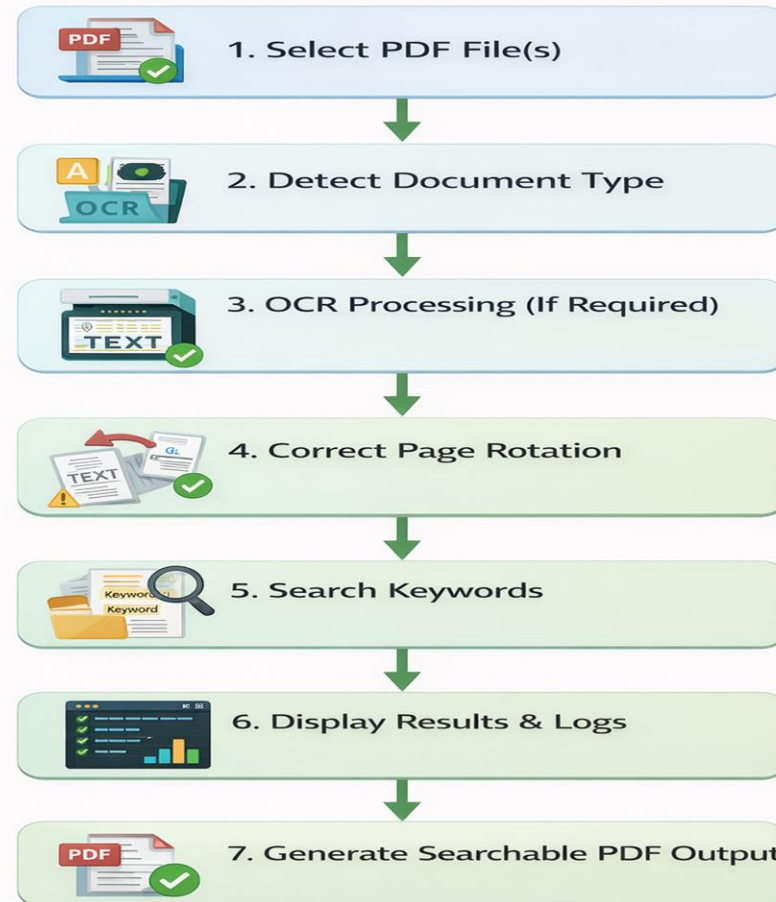


Back-End:

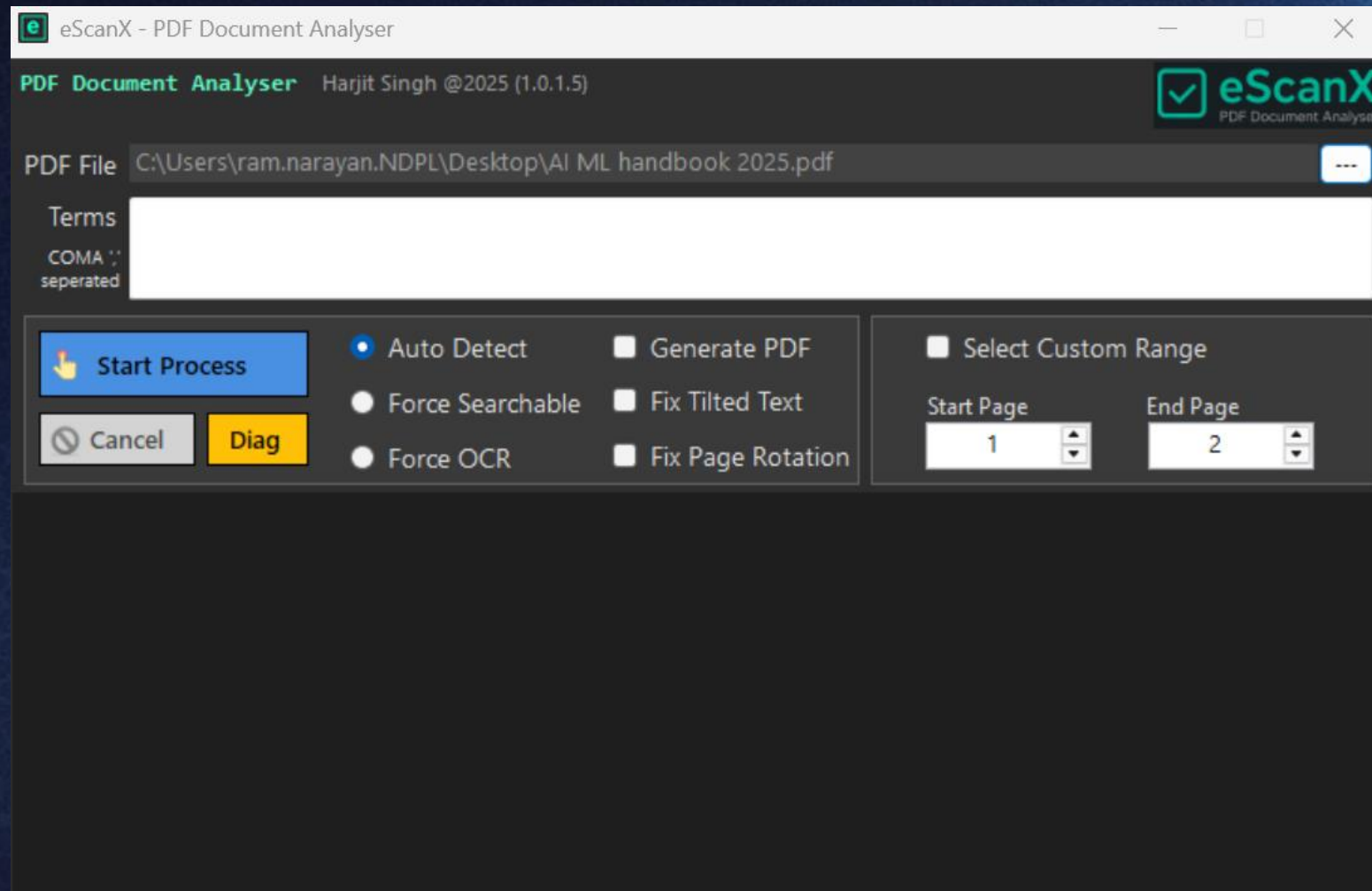
- Python 3.13+ (OCR processing)
- OCR Engine: Tesseract with pytesseract wrapper
- PDF Libraries: pdfplumber, pdf2image, ocrmypdf
- Image Processing: OpenCV & NumPy

Processing Workflow

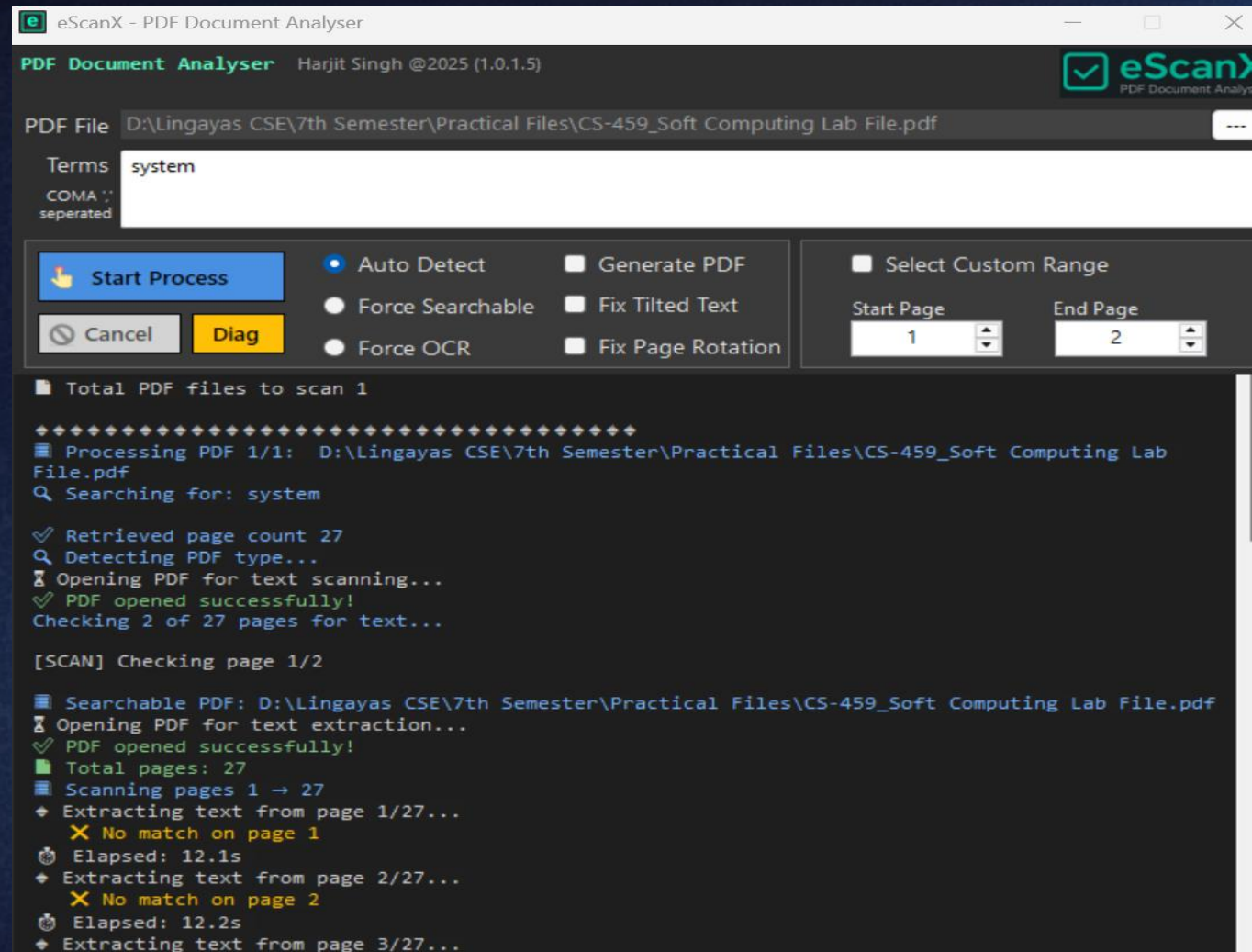
Workflow of eScanX PDF Document Analyzer



Implementation Details (Main Application Interface)



Implementation Details (Search Result Console)



Implémentation Details (OCR Conversion Result Console)

■ Total PDF files to scan 1

```
Processing PDF 1/1: C:\Users\ram.narayan.NDPL\Downloads\ChatGPT Image Dec 16, 2025, 04_17_48 PM.pdf
```

Q Searching for: work

✓ Retrieved page count 1

Q OCR forced...

Scanned PDF: C:\Users\ram.narayan.NDPL\Downloads\ChatGPT Image Dec 16, 2025, 04_17_48 PM.pdf

■ Total pages: 1

Scanning pages 1 → 1

◆ Converting page 1/1...

✓ Match found on page 1

Elapsed: 6.7s

🔍 OCR complete. 1 pages contained search terms.

✓ Found mentions of work:

Page 1:

```
Workflow:of'eScanX PDE:Document: Analyzer:...
```

=====

✓ All PDFs processed successfully.

Performance Metrics

- ❖ **OCR Accuracy:** 98.6% (clear scans) to 84.2% (poor quality)
- ❖ **Keyword Detection:** 95-100% across all document types
- ❖ **Page Rotation Accuracy:** 95% correction success rate
- ❖ **OCR Success Rate:** 98% with zero application crashes

Processing Speed Analysis

- ❖ **Searchable PDF:** 0.3 sec/page
- ❖ **Scanned PDF (OCR):** 2.1 sec/page
- ❖ **Selective scanning significantly reduces total processing time**

Key Advantages

- ✓ Unified solution for all PDF document types
- ✓ Complete offline processing and data privacy
- ✓ Easy installation via MSI setup package
- ✓ Professional UI with minimal learning curve

Applications

- ✓ Academic research and project evaluation
- ✓ Technical manual and SOP analysis
- ✓ Office record digitization

Future Enhancements

- ✓ Cross-platform support (Linux & macOS)
- ✓ Multi-language OCR capabilities
- ✓ AI-based text correction and summarization
- ✓ Cloud integration for large-scale processing

Conclusion

eScanX successfully provides an efficient, accurate, and user-friendly solution for comprehensive PDF document analysis.

Suitable for academic and professional applications

Thank You!

Questions?

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